

# Laptop setup notes btrfs subvolumes

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*Thoughts on laptop configurations for ArchLinux in 2023, with `btrfs` and `sway`*

**\*\*Warning\*\***

This post is a little less complete than most, it has been dated to roughly the correct time chronologically.

## Background

I split this post from the [previous one](#), to make each more self contained. Since the installation media has more features one might perhaps use `gparted` for resizing disks, use the GUI for managing networks, etc. Personally I generally allocate a chunk from whatever other OS is present in `gparted`, and format it as unformatted. This is so I can use [encrypted BTRFS subvolumes](#)<sup>[1]</sup>.

## Input variants with `sway`

Instead of setting up a configuration, I typically use:

```
swaymsg input type:keyboard xkb_variant colemak
swaymsg input type:keyboard xkb_layout us
```

## Setup Encryption

```
cryptsetup luksFormat /dev/nvme0n1p5
cryptsetup open /dev/nvme0n1p5 luks
mkfs.btrfs -L ROOT /dev/mapper/luks
```

## BTRFS Layout

The logic behind the flat `btrfs` layout is to separate things which should remain untouched during rollbacks of the system. I use a variant of a layout from [this issue](#) (more on layouts [here](#) and [here](#)):

```

mount /dev/mapper/luks /mnt
# Root
btrfs sub create /mnt/@ # /
# Needs additional setup
btrfs sub create /mnt/@swap
# Manage each home separately
btrfs sub create /mnt/@home # /home
btrfs sub create /mnt/@pkg # /var/cache/pacman/pkg
btrfs sub create /mnt/@.snapshots
# VMs shouldn't be rolled back
btrfs sub create /mnt/@VMs # /var/lib/libvirt/images
btrfs sub create /mnt/@containers # for docker
# Log files should be consistent
btrfs sub create /mnt/@log # /var/log
# Often has licensed software
btrfs sub create /mnt/@opt # /opt
btrfs sub create /mnt/@usr_local # /usr/local
umount /mnt

```

These can then be mounted with reasonable defaults ([detailed here](#)):

```

mount -o noatime,nodiratime,compress=zstd,space_cache=v2,ssd,subvol=@ /dev/mapper/luks /mnt
mkdir -p /mnt/btrfs # To mount the btrfs root: https://unix.stackexchange.com/q/367123/246530
mount -o noatime,nodiratime,compress=zstd,space_cache=v2,ssd,subvol=id=5 /dev/mapper/luks /mnt/btrfs
# Mount the other subvolumes
mkdir -p /mnt/{boot,home,var/cache/pacman/pkg,.snapshots,var/lib/libvirt/images,var/log,opt,usr/local}
mount -o noatime,nodiratime,compress=zstd,space_cache=v2,ssd,subvol=@home /dev/mapper/luks /mnt/home
mount -o noatime,nodiratime,compress=zstd,space_cache=v2,ssd,subvol=@pkg /dev/mapper/luks
/mnt/var/cache/pacman/pkg
mount -o noatime,nodiratime,compress=zstd,space_cache=v2,ssd,subvol=@VMs /dev/mapper/luks
/mnt/var/lib/libvirt/images
mount -o noatime,nodiratime,compress=zstd,space_cache=v2,ssd,subvol=@log /dev/mapper/luks /mnt/var/log
mount -o noatime,nodiratime,compress=zstd,space_cache=v2,ssd,subvol=@opt /dev/mapper/luks /mnt/opt
mount -o noatime,nodiratime,compress=zstd,space_cache=v2,ssd,subvol=@usr_local /dev/mapper/luks /mnt/usr/local
mount -o noatime,nodiratime,compress=zstd,space_cache=v2,ssd,subvol=@.snapshots /dev/mapper/luks
/mnt/.snapshots

```

We also need to mount the EFI partition:

```

mount -o rw,relatime,fmask=0022,dmask=0022,iocharset=ascii,shortname=mixed /dev/nvme0n1p1 /mnt/boot

```

Switch to sddm with weston for a better display manager

### Handling swap

To get the physical offset, as [noted on the ArchWiki](#) we can use `btrfs inspect-internal [^fn:2]`.

```

cd /mnt/btrfs/@swap
truncate -s 0 ./swapfile
chattr +C ./swapfile
btrfs property set ./compression none
dd if=/dev/zero of=./swapfile bs=1M count=<FILE-SIZE-IN-MiB> status=progress
chmod 600 ./swapfile
mkswap ./swapfile
swapon ./swapfile
cd -

```

## System setup

I like the live USB packages, so I tend to use the `./offlineInst.sh` instead of `pacstrap`. Some additional configuration is needed:

```
arch-chroot /mnt
# Set the hostname
echo <HOSTNAME> > /etc/hostname
# Locale is set already to C.UTF-8
# Setup timezone
ln -sf /usr/share/zoneinfo/<WHATEVER> /etc/localtime

# Hosts should be updated
# cat /etc/hosts
# #<ip> <hostname.domain.org> <hostname>
# 127.0.0.1 <HOSTNAME>.localdomain <HOSTNAME>
# ::1 localhost.localdomain localhost
```

Finally, a password needs to be setup for the `root` user with `passwd`. It is generally a good idea to make a new user too:

```
useradd -m username -s /bin/zsh
passwd username
usermod -aG wheel username # for sudo
```

Some other (optional) `colemak` steps: `#+begin_src bash echo KEYMAP=colemak > /etc/vconsole.conf`  
`#+end_src`

## Cleaning up `archiso`

Some cruft is copied over if the `offlineInst.sh` script is used. We will need to:

- Remove the local repository from `/etc/pacman.conf`
- Remove the preset `archiso` for `mkinitcpio` (by deleting `/etc/mkinitcpio.d/linux.preset`)

We also need to remove the user:

```
userdel hzlinarch
rm -rf /home/hzlinarch
```

Some other things to be deleted (should be updated in the script):

```
sudo rm -rf /var/lib/pacman/local/mkinitcpio-archiso-*
sudo rm -rf /usr/lib/initcpio/hooks/archiso*
sudo rm -rf /usr/share/doc/mkinitcpio-archiso
```

## Kernel setup

The `initramfs` hooks need to be modified for `swap` and to recognize `btrfs` (also to remove all the `archiso` hooks):

```
# /etc/mkinitcpio.conf
HOOKS=(base keyboard udev kms autodetect modconf block keymap encrypt btrfs filesystems resume)
```

Additionally, the kernel will need to be recompiled (at the very least). Since I was using the `nitrous` variant, this time I recompiled with `mperformance` in the `PKGBUILD` and this meant modifying the `rtl88xxau` kernel driver a bit as it passes `-mhard-float` and `clang` will fail with unused arguments.

## Setup notes

With the skeletal system setup, I needed only to add my customizations. Namely:

- All `clang` source builds
- SDDM display management

## Nix

My dotfiles expect `nix` for a variety of additional reasons. With `sddm` however, the default installation adds all `nixbld` users to the login screen. To prevent this, it is best to set the `MaximumUid`, for example in `/etc/sddm.conf.d/08_nix_uid.conf`:

```
[Users]
MaximumUid=30000
```

## Wayland notes

- `caja` isn't compatible with `wayland` out of the box (but my favorite `thunar` is)
- Thanks to a [very old](#) archive-manager (file-roller) bug, there aren't any really good native options for compression
  - `peazip` is a native file and archive manager though
- `kdiallog` turned out to be really useful for handling issues with Chromium browsers
  - Wasn't able to print or attach files or anything otherwise

## Laptop Tweaks

Mostly garnered from the [Archwiki Laptop page](#). I generally setup `/etc/systemd/{logind,sleep}.conf` with defaults from my older `etckeeper` configuration.

- Fingerprints worked natively with `fprintd` and a polkit (I used `lxpolkit`)
- For keyring management `gnome-keyring` is fine (with `wl-clipboard`)

## Conclusions

At this point, my notes came to an abrupt end. Presumably this meant I was able to cobble together the rest of the system using the pointers there.